

Is interval training the magic bullet for fat loss? A systematic review and meta-analysis comparing moderate-intensity continuous training with high-intensity interval training (HIIT)

Ricardo Borges Viana,¹ João Pedro Araújo Naves,¹ Victor Silveira Coswig,² Claudio Andre Barbosa de Lira,¹ James Steele,³ James Peter Fisher,³ Paulo Gentil¹

► Additional material is published online only. To view please visit the journal online (<http://dx.doi.org/10.1136/bjsports-2018-099928>).

¹Faculty of Physical Education and Dance, Federal University of Goiás, Goiânia, Goiás, Brazil

²Faculty of Physical Education, Federal University of Pará, Castanhal, Pará, Brazil

³Centre for Health, Exercise, and Sport Science, School of Sport, Health and Social Sciences, Southampton, Hampshire, UK

Correspondence to

Dr Paulo Gentil, Faculdade de Educação Física e Dança, Universidade Federal de Goiás, Goiânia 74605-220, Brazil; paulogentil@hotmail.com

Accepted 14 December 2018
Published Online First
14 February 2019



© Author(s) (or their employer(s)) 2019. No commercial re-use. See rights and permissions. Published by BMJ.

To cite: Viana RB, Naves JPA, Coswig VS, et al. *Br J Sports Med* 2019;**53**:655–664.

ABSTRACT

Objectives To compare the effects of interval training and moderate-intensity continuous training (MOD) on body adiposity in humans, and to perform subgroup analyses that consider the type and duration of interval training in different groups.

Design Systematic review and meta-analysis.

Data sources English-language, Spanish-language and Portuguese-language searches of the electronic databases PubMed and Scopus were conducted from inception to 11 December 2017.

Eligibility criteria for selecting studies Studies that met the following criteria were included: (1) original articles, (2) human trials, (3) minimum exercise training duration of 4 weeks, and (4) directly or indirectly compared interval training with MOD as the primary or secondary aim.

Results Of the 786 studies found, 41 and 36 were included in the qualitative analysis and meta-analysis, respectively. Within-group analyses showed significant reductions in total body fat percentage (%) (interval training: -1.50 [95% CI -2.14 to -0.86 , $p<0.00001$] and MOD: -1.44 [95% CI -2.00 to -0.89 , $p<0.00001$]) and in total absolute fat mass (kg) (interval training: -1.58 [95% CI -2.74 to -0.43 , $p=0.007$] and MOD: -1.13 [95% CI -2.18 to -0.08 , $p=0.04$]), with no significant differences between interval training and MOD for total body fat percentage reduction (-0.23 [95% CI -1.43 to 0.97], $p=0.705$). However, there was a significant difference between the groups in total absolute fat mass (kg) reduction (-2.28 [95% CI -4.00 to -0.56], $p=0.0094$). Subgroup analyses comparing sprint interval training (SIT) with MOD protocols favour SIT for loss of total absolute fat mass (kg) (-3.22 [95% CI -5.71 to -0.73], $p=0.01$). Supervised training, walking/running/jogging, age (<30 years), study quality and intervention duration (<12 weeks) favourably influence the decreases in total absolute fat mass (kg) observed from interval training programmes; however, no significant effect was found on total body fat percentage (%). No effect of sex or body mass index was observed on total absolute fat mass (kg) or total body fat percentage (%).

Conclusion Interval training and MOD both reduce body fat percentage (%). Interval training provided 28.5% greater reductions in total absolute fat mass (kg) than MOD.

Trial registration number CRD42018089427.

INTRODUCTION

Whether physical activity affects weight control has been an ongoing topic of controversy.^{1,2} The majority of physical activity guidelines for the management of obesity recommend high exercise volumes.^{3,4} Guidelines generally recommend 150–250 min/week, and up to 60 min/day, of moderate-intensity aerobic exercise to prevent weight gain or to reduce body mass a little bit (2–3 kg).^{4,5} More than an hour of exercise daily (>420 min/week) is recommended to lose more weight (5–7.5 kg)³ and few people meet these guidelines.^{6,7}

Interval training may have the potential to promote weight loss as it has some benefits similar to moderate-intensity continuous training (MOD) while requiring less time.^{8,9} MOD is typically defined as continuous effort that elicits 55%–70% of the maximal heart rate (HRmax) or promotes oxygen consumption (VO_2) equivalent to 40%–60% of the maximum VO_2 ($\text{VO}_{2\text{max}}$).¹⁰ Interval training is an intermittent period of effort interspersed by recovery periods¹¹; the two most common types of interval training are high-intensity interval training (HIIT) and sprint interval training (SIT).⁷ HIIT requires ‘near maximal’ efforts performed at a heart rate (HR) $\geq 80\%$ of the HRmax or the equivalent as expressed in the function of the $\text{VO}_{2\text{max}}$. Even more intense exercise, SITs are efforts performed at intensities equal or superior to the one that elicited a peak VO_2 on an incremental test ($i\text{VO}_{2\text{peak}}$), including ‘all-out’ efforts.^{7,12}

HIIT programmes, when compared with MOD, promote greater increases in $\text{VO}_{2\text{max}}$,¹³ ventricular and endothelial function,¹⁴ greater or comparable improvements in insulin sensitivity¹⁵ and blood pressure,¹⁶ lower ratings of perceived exertion,¹⁷ similar⁶ or higher levels of enjoyment,^{17,18} and similar⁶ or higher adherence¹⁸ than MOD, depending on how the programme is designed. In addition, despite lower training volume in SIT programmes, SIT may promote increases in skeletal muscle oxidative capacity,¹⁹ specific metabolic adaptations during exercise¹⁹ and exercise performance similar to MOD.²⁰

Decreases in body fat may be similar²¹ or higher²² in interval training than MOD. Interval training may elicit greater weight loss even if the energy expenditure obtained during the interval training is lower²³ or equal²⁴ to that during MOD. This may

Review

be due to greater resting energy expenditure and fat utilisation immediately following interval training exercise.^{25 26} However, there are currently many different approaches to performing interval training,²⁷ and there is still no consensus as to which training method (HIIT/SIT vs MOD) is best to reduce body fat. The considerable variability among interval training protocols,²⁷ compared with fairly homogeneous MOD protocols, introduce 'noise' in the literature. No study has yet addressed the simple question: Which type of exercise is better for weight loss?

We conducted a systematic review, qualitative appraisal and meta-analysis of studies that directly or indirectly compared the effects of HIIT or SIT with MOD on adiposity. We compared subgroups to test whether (1) the nature of the interval training (HIIT or SIT), (2) sex, (3) baseline body mass index (BMI) or other variables influenced the outcome. We hypothesised that HIIT/SIT would reduce body fat more effectively than MOD.

METHODS

The results of this systematic review and meta-analysis are presented according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses statement,²⁸ and was preregistered in the International Prospective Register of Systematic Review (PROSPERO).²⁹

Search strategy

English-language, Spanish-language and Portuguese-language searches of the electronic databases PubMed and Scopus were conducted from inception to 11 December 2017 by two independent researchers (RBV and JPAN). Articles were retrieved from electronic databases using the following search criteria: (interval training OR intermittent training OR high intensity OR sprint interval training OR aerobic interval training HIIT OR HIIIE OR high intensity interval training OR high-intensity interval training OR high intensity interval exercise OR high-intensity interval exercise OR high intensity intermittent exercise OR high-intensity intermittent exercise OR high intensity intermittent training OR high-intensity intermittent training) AND (continuous training OR moderate-intensity continuous exercise OR moderate intensity continuous exercise OR moderate-intensity continuous training OR moderate intensity continuous training) AND (body fat OR adiposity OR body composition OR abdominal fat OR visceral fat OR adipose tissue) AND Humans.

Initially, titles and abstracts of identified studies were checked for relevance by two reviewers (RBV and JPAN). Subsequently, the reviewers independently reviewed the full text of potentially eligible studies. Any disagreement for inclusion between the reviewers was resolved by a third researcher (PG). Additional studies were identified via hand-searching and reviewing the reference lists of relevant papers. All these steps were performed for 3 weeks. Figure 1 presents the flow of papers through the study selection process.

Inclusion and exclusion criteria: participants, interventions, comparators and outcomes

Studies with participants of all ages and sexes with a minimum exercise training duration of 4 weeks, which directly or indirectly compared HIIT or SIT with MOD as the primary or secondary aim (according to previous definitions), and which evaluated fat change by methods that infer total or regional mass, or total or regional percentage fat, were included. Studies that reported only BMI and that compared HIIT or SIT or MOD with only non-training control groups were not included for analysis. When employing two interval training protocols, both were included,

in different analyses, for comparison with MOD. Studies were excluded based on the following article types: letters to the editor, books, book sections, theses, film/broadcasts, opinion articles, observational studies and abstracts without adequate data, or reviews.

Data extraction

The following study characteristics were extracted: age, sex, body mass, BMI, $\text{VO}_{2\text{max/peak}}$, total or regional fat mass (kg), percentage total and regional body fat (%), and HIIT/SIT and MOD interventions characteristics. These data were extracted independently by two researchers (RBV and JPAN), with disagreements resolved by a third researcher (PG). When studies provided insufficient data for inclusion in the meta-analysis (five studies), the corresponding authors were contacted via email to determine whether additional data could be provided; however, no corresponding authors responded.

Study quality assessment

Study quality was assessed by two researchers (RBV and JPAN) using a modified Downs and Black checklist.³⁰ Items included the appropriate reporting of the hypotheses, outcomes, interventions, adverse events, participant characteristics (a clear statement on inclusion and exclusion criteria), descriptions of patients lost to follow-up (studies with $\geq 10\%$ dropout without characteristics reported scored 0), assessment method accuracy, statistical methods, blinding and randomisation procedures. The scale was modified to include criteria for monitoring and reporting of physical activity level (yes=1, no=0) and diet (yes=1, no=0), the supervision of exercise sessions (yes=1, no=0), and information about adherence and/or compliance to exercise interventions (yes=1, no=0). Therefore, the studies that monitored and reported control of diet, habitual activity, supervision, and adherence or compliance scored 1 point in each item. If an item was unable to be determined, it was scored as 0. The highest possible score for quality was 20. In addition, we recorded the strengths/weakness/unknowns of available information from studies to strengthen the quality analysis of the included studies.

Statistical analyses

All analyses were conducted using the R package (V.3.2.4). Meta-analysis was conducted using a random-effects model (DerSimonian and Laird approach) for the individual effects of HIIT/SIT and MOD on total body fat (kg) and body fat percentage. The random-effects model was preferred to a fixed-effect model as certain experimental parameters had wide variation. For the secondary meta-analysis, premeans, postmeans, absolute and relative changes, and SD for each group were collected. Initially, a within-group effect size (ES) was calculated using a random-effects model (DerSimonian and Laird approach) to estimate the change from baseline for each group, given that a random-effects model considers true random errors within a single study and variation in effects occurring from study to study. The statistical heterogeneity of the treatment effect among studies was assessed using Cochran's Q test and the inconsistency I^2 test, in which values above 30% and 50% were considered indicative of moderate and high heterogeneity, respectively.³¹ Publication bias was assessed with funnel plots and Begg's test. To improve our results, we conducted several sensitivity analyses to consider the individual influence of each study on the overall results, as well as the type of comparison group (HIIT or SIT), type of modality (walking/jogging/running or cycling), age (<30 and ≥ 30 years), sex, BMI ($<30 \text{ kg/m}^2$ and $\geq 30 \text{ kg/m}^2$), intervention duration

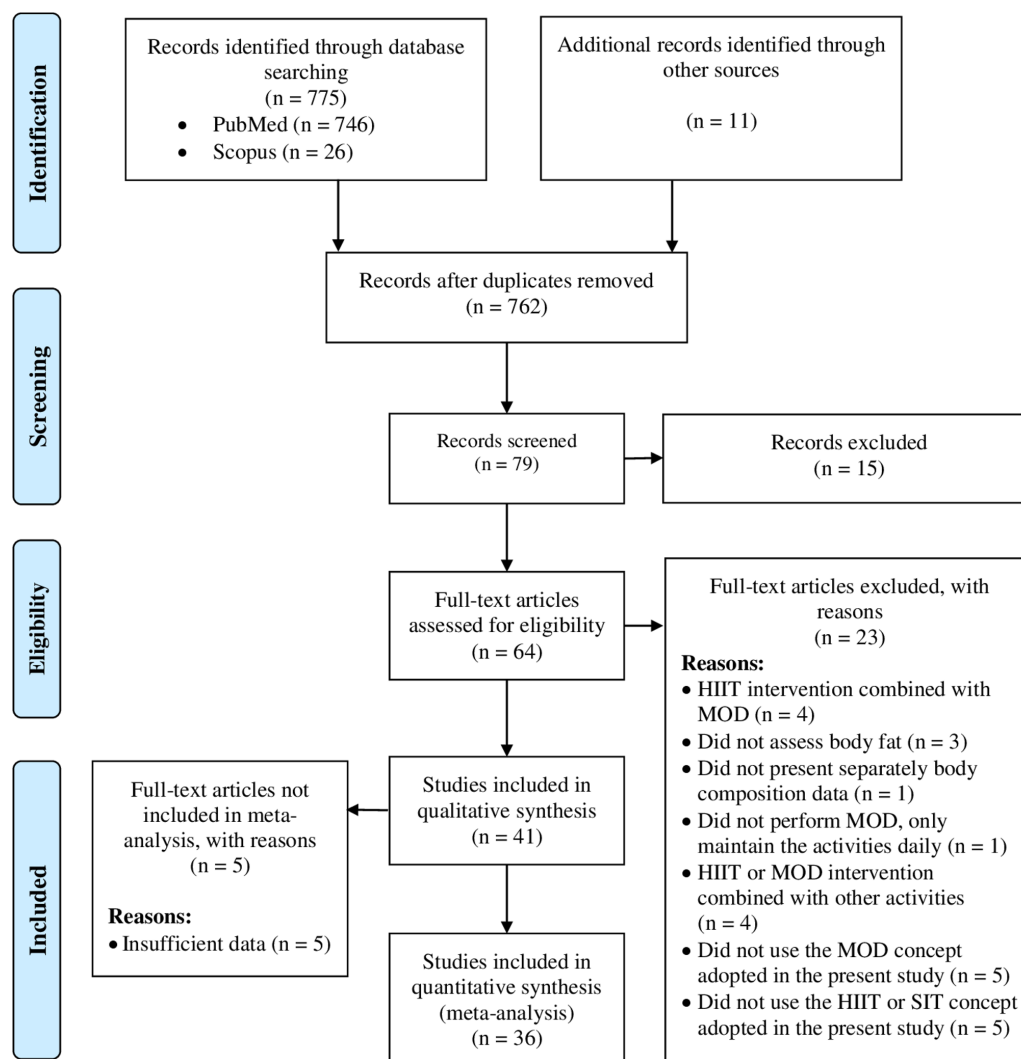


Figure 1 Flow diagram of outcomes of review. HIIT, high-intensity interval training; MOD, moderate-intensity continuous training; SIT, sprint interval training.

(<12 and ≥ 12 weeks), study quality ('low, middle and high') and supervision of exercise sessions (yes and no).

RESULTS

Included studies

The search strategy retrieved 786 records. After deduplication and language examination (English, Spanish and Portuguese studies), 24 studies were excluded from the review process and 698 were excluded after title and/or abstract analysis; 64 full-text copies of the remaining studies were obtained and subjected to further evaluation. After reading full-text copies, 23 studies were excluded from this review due to the following reasons: (1) four studies included MOD in combination with the HIIT protocol^{23 32–34}; (2) one study did not perform MOD³⁵; (3) four studies applied an HIIT or MOD intervention combined with other activities^{36–39}; (4) five studies did not use the MOD criteria adopted in this review^{40–44}; (5) five studies stated the use of HIIT or SIT protocols,^{45–49} but did not match the HIIT and SIT criteria adopted in this review; (6) three studies did not assess body fat^{50–52}; and (7) one study did not present separate body composition data.⁵³ At the end of the process, 41 publications meeting the eligibility criteria were included for qualitative analyses,^{22 52–91} of which 5 studies provided insufficient data and

were excluded from this review due to the following reasons: (1) two studies did not provide SD for change in mean values^{67 92}; (2) two studies reported only regional body fat percentage (%)^{58 84}; and (3) one study provided only skinfold values.⁵⁵ Therefore, 36 studies provided sufficient data for meta-analysis (35 for total body fat percentage [%] and 15 for total absolute fat mass [kg]) (figure 1).

Participant characteristics

Participants' characteristics are summarised in table 1. Overall, 1115 participants were included in the qualitative analysis and 1012 in the meta-analysis. The number of participants in the studies varied from 7⁶⁸ to 90.⁷⁸ Fourteen studies examined exclusively males,^{55 59 66 67 69 72 73 75 77 79–81 88 92} nine exclusively females,^{24 63 65 74 82 83 87 90 93} one did not report the number of males and females used to present the body composition results,⁸⁴ while the remaining studies (n=17) assessed a mixed-sex sample.^{44 54 56–58 60–62 64 68 70 71 76 78 85 86 89 91} In total, 576 males and 522 females participated in the studies. Two studies used the same sample,^{61 62} and one performed a double-blind, randomised, crossover investigation with seven athletes (five males and two females).⁶⁸ The mean age of study participants ranged from 10.4⁶⁴ to 70.1 years.⁸⁷ The training status of

Review

Table 1 Characteristics of the participants

Reference	Participants*	Male (%)/ Female (%)	Age (year)	BMI (kg/m ²)	Other population characteristics
Thomas <i>et al</i> ⁵⁴	29	38/62	18–32	NR	Healthy, untrained but active young adults
Mäder <i>et al</i> ⁵⁵	14	100/0	HIIT: 28–46 MOD: 28–46	HIIT: 25.9 (3.2) MOD: 26.9 (3.3)	Untrained but slightly overweight
Trapp <i>et al</i> ²⁴	30	0/100	20.2 (7.7)	23.2 (7.7)	Healthy, untrained young
Schjerve <i>et al</i> ⁵⁶	27	22/78	HIIT: 46.9 (8.2) MOD: 44.4 (7.6)	HIIT: 36.6 (5.5) MOD: 36.7 (5.1)	Adults with obesity
Moreira <i>et al</i> ⁵⁷	16	36/64	40 (8)	HIIT: 28.3 (3.7) MOD: 27.5 (1.9)	Healthy, overweight
Wallman <i>et al</i> ⁵⁸	13	31/69	HIIT: 40.9 (11.7) MOD: 44.8 (16.8)	HIIT: 31.4 (2.6) MOD: 30.1 (2.6)	Overweight/obese adults
Nybo <i>et al</i> ⁵⁹	17	100/0	HIIT: 37.0 (8.5) MOD: 31.0 (6.0)	NR	Healthy untrained
Macpherson <i>et al</i> ⁶⁰	20	40/60	24.0 (3.0)	NR	Healthy, recreationally active university students
Buchan <i>et al</i> ⁶¹	33	82/18	SIT: 16.7 (0.1) MOD: 16.2 (0.1)	SIT: 21.6 (2.2) MOD: 22.4 (3.3)	Adolescents
Buchan <i>et al</i> ⁶²	33	82/18	SIT: 16.7 (0.1) MOD: 16.2 (0.1)	SIT: 21.6 (2.2) MOD: 22.4 (3.3)	Adolescents
Sijie <i>et al</i> ⁶³	33	0/100	HIIT: 19.8 (1.0) MOD: 19.3 (0.7)	HIIT: 27.7 (1.9) MOD: 28.3 (2.0)	Overweight/obese young
Corte de Araujo <i>et al</i> ⁶⁴	30	30/70	HIIT: 10.4 (0.9) MOD: 10.7 (0.7)	HIIT: 32.0 (3.0) MOD: 30.0 (4.0)	Children with obesity
Eimarieskandari <i>et al</i> ⁶⁵	14	0/100	HIIT: 22.3 (0.9) MOD: 21.4 (0.5)	HIIT: 29.2 (0.8) MOD: 30.7 (2.3)	Young adults with obesity
Koubaa <i>et al</i> ⁶⁶	29	100/0	HIIT: 13.0 (0.8) MOD: 12.9 (0.5)	HIIT: 30.2 (3.6) MOD: 30.8 (2.9)	Adolescent boys with obesity
Earnest <i>et al</i> ⁶⁷	37	100/0	HIIT: 48 (9) MOD: 49 (9)	HIIT: 30.4 (2.3) MOD: 31.4 (3.4)	Adults
Shing <i>et al</i> ⁶⁸ §	7	71/29	19.0 (1.2)	NR	Junior state-level and national-level rowers
Shepherd <i>et al</i> ⁶⁹	16	100/0	SIT: 22.0 (2.8) MOD: 21.0 (2.8)	SIT: 24.8 (2.3) MOD: 22.6 (4.5)	Healthy and inactive adults
Keating <i>et al</i> ⁷⁰	22	23/77	HIIT: 41.8 (2.7) MOD: 44.1 (1.9)	HIIT: 28.2 (0.5) MOD: 28.5 (0.6)	Inactive, overweight adults
Lunt <i>et al</i> ⁷¹	49	27/73	HIIT: 48.2 (5.6) MOD: 46.3 (5.4) SIT: 50.3 (8.0)	HIIT: 32.1 (3.1) MOD: 32.7 (3.4) SIT: 32.4 (2.9)	Inactive, overweight/obese adults
Nalcakan ⁷²	15	100/0	21.7 (2.2)	SIT: 25.5 (2.2) MOD: 24.5 (1.9)	Healthy and young recreationally active university students
Sasaki <i>et al</i> ⁷³	24	100/0	NR	HIIT: 24.3 (0.7) MOD: 23.4 (0.8)	Healthy and sedentary
Mohr <i>et al</i> ⁷⁴	42	0/100	SIT: 44 (2) MOD: 46 (2)	>25.0	Sedentary premenopausal women with mild to moderate arterial hypertension
Cocks <i>et al</i> ⁷⁵	16	100/0	25.0 (2.8)	34.8 (0.9)	Inactive young with obesity
Cheema <i>et al</i> ⁷⁶	12	58/42	HIIT: 43 (19) MOD: 36 (15)	HIIT: 32.0 (5.9) MOD: 30.8 (2.6)	Inactive adults with central obesity
Elmer <i>et al</i> ⁷⁷	12	100/0	HIIT: 21.4 (1.1) MOD: 21.8 (2.1)	HIIT: 24.7 (2.9) MOD: 27.1 (4.8)	Healthy sedentary or inactive adults
Shepherd <i>et al</i> ⁷⁸	90	33/67	HIIT: 42.0 (11) MOD: 43 (11)	HIIT: 27.7 (5.0) MOD: 27.7 (4.6)	Healthy and inactive adults
Fisher <i>et al</i> ⁷⁹	23	100/0	HIIT: 20.0 (1.5) MOD: 20.0 (1.5)	HIIT: 30.0 (3.1) MOD: 29.0 (3.4)	Inactive, overweight/obese young men
Sim <i>et al</i> ⁸⁰	20	100/0	31.8 (8.0)	HIIT: 27.4 (1.6) MOD: 27.2 (1.5)	Inactive, overweight/obese adult participants
Devin <i>et al</i> ⁸¹	35	100/0	HIIT: 61.4 (11.1) MOD: 61.5 (10.8)	HIIT: 27.1 (4.8) MOD: 26.4 (3.4)	Colorectal cancer survivors
Zhang <i>et al</i> ⁸²	24	0/100	HIIT: 21.0 (1.0) MOD: 20.6 (1.2)	HIIT: 25.8 (2.7) MOD: 26.0 (1.6)	Chinese ethnicity, inactive, overweight/obese
Gillen <i>et al</i> ⁸³	18	0/100	SIT: 27 (7) MOD: 28 (9)	SIT: 27 (5) MOD: 26 (6)	Inactive
Martins <i>et al</i> ⁸⁴	17†	NR	SIT: 33.9 (7.8) 1/2SIT: 34.1 (7.1) MOD: 33.0 (9.9)	SIT: 33.2 (3.5) 1/2SIT: 32.4 (2.9) MOD: 33.3 (2.4)	Inactive adults with obesity

Continued

Table 1 Continued

Reference	Participants*	Male (%)/ Female (%)	Age (year)	BMI (kg/m ²)	Other population characteristics
Hwang <i>et al</i> ⁸⁵	29	41/59	65 (7.1)	HIIT: 28.0 (4.3) MOD: 28.7 (3.7)	Inactive and healthy older adults
Ramos <i>et al</i> ⁸⁶	66	63/37	4HIIT: 56 (10) 1HIIT: 58 (7) MOD: 57 (9)	4HIIT: NR 1HIIT: NR MOD: NR	Adults with metabolic syndrome
Maillard <i>et al</i> ⁸⁷	16	0/100	HIIT: 68.2 (1.9) MOD: 70.1 (2.4)	HIIT: 32.6 (1.7) MOD: 29.7 (1.2)	Postmenopausal and obese (61–81 years) women with type 2 diabetes
Higgins <i>et al</i> ⁸⁸	52	100/0	20.4 (1.5)	30.3 (4.5)	Inactive, overweight/obese women
Boer and Moss ⁸⁹	26	58/42	SIT: 30.0 (7.0) MOD: 34.2 (9.2)	SIT: 29.3 (4.0) MOD: 30.6 (6.1)	Adults with Down syndrome
Panissa <i>et al</i> ⁹⁰	23	0/100	HIIT: 30.6 (15.1) MOD: 26.1 (9.1)	HIIT: 25.9 (4.1) MOD: 23.3 (2.3)	Inactive healthy women
Camacho-Cardenosa <i>et al</i> ⁹¹	34	54/46†	SIT: 11.1 (0.2) MOD: 11.3 (0.5)	SIT: 18.4 (2.8) MOD: 20.1 (3.3)	Adolescents
Zhang <i>et al</i> ⁹³	30	0/100	HIIT: 21.5 (1.7) MOD: 20.9 (1.4)	≥25	Obese young women
Galedari <i>et al</i> ^{92†}	22	100/0	HIIT: 30.8 (7.6) MOD: 28.8 (6.1)	HIIT: 29.6 (1.5) MOD: 28.9 (1.3)	Overweight men

*Number included in HIIT/SIT versus MOD for body composition analysis.

†n=17 for body composition analysis in HIIT versus MOD.

‡Percentage referring to 35 participants. Values reported as mean.

§Double-blind, randomised, crossover investigation.

1/2SIT, approximately half of time of the sprint interval training; 1HIIT, high-intensity interval training (one bout); 4HIIT, high-intensity interval training (four bouts); BMI, body mass index; HIIT, high-intensity interval training; MOD, moderate-intensity continuous training; NR, not reported; SIT, sprint interval training.

the participants ranged from sedentary⁷³ to high-level athletes.⁶⁸ The mean BMI ranged from 18.4 kg/m²²⁹¹ to 36.7 kg/m².⁵⁶

Intervention characteristics

The interval training and MOD programmes are summarised in online supplementary table S1. According to the criteria of HIIT and SIT adopted in this review,^{71 25}^{54–59 63–68 70 73 76–78 81 82 85–87 90 92 93} and 15^{24 60–62 69 72 74 75 79 80 83 84 88 89 91} of the 41 included studies employed HIIT and SIT interventions, respectively. Only one study⁷¹ employed both HIIT and SIT interventions, and one study included two HIIT interventions.⁸⁶ Of the 41 studies, 20 used cycling,^{24 55 57 58 69 70 72 73 75 78–81 83 84 87–90 93} 16 used walking/jogging/running,^{54 56 59–67 71 77 82 91 92} 1 used a synchronous arm and leg air-braked ergometer,⁸⁵ 1 offered a choice between the two (cycle ergometer or walking/running) depending on orthopaedic limitations,⁸⁶ 1 used swimming,⁷⁴ 1 used boxing drills for the HIIT protocol and walking for MOD,⁷⁶ and 1 used a rower ergometer.⁶⁸ Intervention duration ranged from 4^{68 73 75 81} to 16 weeks,^{86 87} with 12 weeks being the most common (~44%; n=18)^{54 56 57 59 63 64 66 67 70 71 76 80 82–84 89 92 93} (online supplementary table S1). The most widely used HIIT (n=8) protocol consisted of alternating 4 min at high intensity followed by 3 min of recovery.^{56 65 71 81 82 85 86 93} The most widely used SIT protocols consisted of alternating 30 s 'all-out' efforts followed by 4 min of recovery,^{60 79 88} and protocols that alternate 8 s 'all-out' efforts followed by 12 s of recovery.^{24 84} The MOD protocols used lasted from 10²⁴ to 60 min,^{59 74} with 40–45 min (n=6)^{63 65 70 76 83 87} and 29–35 min (n=6)^{71 77 82 85 86 90} being the most used protocols.

Twenty-two HIIT protocols used active recovery,^{54–56 58 63–65 67 68 70 76–78 81 82 84–87 89 90 92} one used passive recovery,⁹³ and five did not report clearly what type of recovery was used.^{57 59 66 73 91} Eight SIT protocols used active recovery,^{54 60 69 75 79 80 83 88} one used passive recovery,⁷⁴ and three did not report clearly what type of recovery was used.^{61 62 72} The only study to employ HIIT versus SIT versus MOD protocols⁷¹ used active recovery in both the interval training protocols.

The intensity of effort for HIIT protocols was prescribed by the percentage of $i\dot{V}O_{2\max}$ ^{66 67 77} or $i\dot{V}O_{2\text{peak}}$,⁵⁸ percentage of $\dot{V}O_{2\max}$ ^{55 63 73} or $\dot{V}O_{2\text{peak}}$,⁷⁰ percentage of HRmax^{54 56 59 71 78 87 90 92 93} or peak heart rate (HRpeak),^{64 65 81 82 85 86} rating of perceived exertion,⁷⁶ HR corresponding to 20% above the HR at ventilatory threshold,⁵⁷ and 90% of 4 min maximal power.⁶⁸ The intensity of effort in most SIT protocols (n=13) was prescribed by 'all-out' efforts,^{24 60–62 69 71 72 74 83 84 88 89 91} percentage of $i\dot{V}O_{2\text{peak}}$,⁸⁰ percentage of maximal power output⁷⁵ and percentage of anaerobic power.⁷⁹

More than half of the protocols (~63%; n=26) were performed three times per week.^{24 54–57 59–62 65 66 70–74 77 80 81 83 84 88–92} Four protocols were performed four times per week,^{58 76 82 85} two protocols were performed three to four times per week,^{67 93} three protocols were performed twice a week,^{64 68 87} one protocol was performed five times per week,⁶³ and five MOD protocols had a frequency greater (five times per week) than the interval training protocols (three times per week).^{69 75 78 79 86}

Diet and physical activity control

Almost half of the studies (~42%; n=17) instructed participants to maintain both normal diet and physical activity.^{24 60 61 63 69 70 72 73 77 78 81 84 85 87 90 91 93} Twenty-three (~56%) and 19 (~46%) studies reported a diet^{24 55 58 61 62 64 65 67–71 73 80 82 84 86–88 90–93} and physical activity control,^{56 58 62 67 68 70 71 73 78 80–82 84–88 91 93} respectively (online supplementary table S2). One study provided a 1-hour diet education session per week.⁵⁸ In addition, one study employed a caloric reduction of 500 kcal/day based on participants' normal intake.⁹² Online supplementary table S1 shows additional information about diet and physical activity of the participants of the included studies.

Body composition assessments

Most studies (~56%; n=23) used only dual-energy X-ray absorptiometry (DXA) to determine total and/or android, trunk

and gynoid body composition.^{24 56 58 59 63 67–71 74 75 77 79–81 84–88 92 93} Others used bioelectrical impedance,^{57 64 65 78 89 91} hydrodensitometry,⁵⁴ air displacement plethysmography^{60 83} or skinfold measurements.^{61 62 66 72 76 90} Three studies used two body composition assessments methods, such as DXA and CT,⁸⁷ bioelectrical impedance and CT,⁸² and bioelectrical impedance and MRI.⁷³ Online supplementary table S3 shows additional information about body composition assessment methods used in the included studies.

Quality assessment

A modified Downs and Black checklist³⁰ assessment determined that the quality of studies had a mean score of 13.5 ± 2.3 (ranging from 9⁵⁴ to 19⁷⁰; online supplementary table S1). All included studies specified their main findings and outcomes, participant characteristics, statistical tests and accurate measures. Only one study⁸³ did not report variability estimates. No studies blinded participants to exercise intervention, and only eight (~20%) blinded assessors to group allocation.^{67 68 70 71 76 82 85 86} Most studies (~88%; n=36) randomised participants to groups.^{24 54–58 60–64 66–71 73–76 78–82 84–93} Twenty-five studies (~61%) reported adherence or compliance.^{56 59 61 62 64 67 68 70–72 74 76–78 80–86 88–90 93} Nineteen studies (~46%) adequately reported adverse events.^{57 59–64 70–72 76 77 79 81 82 85 88 89 93} Four studies (~10%) did not provide information about supervision of exercise sessions.^{72 73 75 76}

Meta-analysis

The within-group analysis found that interval training (−1.50 [95% CI −2.14 to −0.86, $p=0.00001$]) and MOD (−1.44 [95% CI −2.00 to −0.89, $p<0.0001$]) resulted in significant improvements in total body fat percentage (%) (online supplementary figures S1A and S2A, respectively). Significant improvements also were found in total absolute fat mass (kg) for HIIT/SIT (−1.58 [95% CI −2.74 to −0.43, $p=0.007$]) and MOD (−1.13 [95% CI −2.18 to −0.08, $p=0.04$]) (online supplementary figures S1B and S2B, respectively).

Primary analysis

The between-group analyses on the effects of interval training versus MOD on total body fat percentage (%) and total absolute fat mass (kg) are presented in figure 2 and figure 3, respectively. Overall, there was no difference between groups in total body fat percentage (%) ($p=0.705$), with evidence of significant heterogeneity in the meta-analysis of total body fat percentage (%) ($I^2=75.4\%$, $p<0.0001$). However, there was a significant difference between groups in total absolute fat mass (kg) ($p=0.0094$), favouring interval training, with evidence of significant heterogeneity in the meta-analysis of total absolute fat mass (kg) ($I^2=48.4\%$, $p=0.0184$).

Subgroup analyses demonstrated a significant effect of interval training mode (SIT vs MOD), modality of exercise (walking/jogging/running vs cycling), supervision (yes vs no), study quality (low vs middle vs high), age (<30 vs ≥ 30 years) and intervention duration (<12 vs ≥ 12 weeks) on total absolute fat mass (kg) (online supplementary figures S3–S14); however, no significant effect was found on total body fat percentage (%). No effect of sex or BMI was observed on total absolute fat mass (kg) or total body fat percentage (%) (online supplementary figures S15–S18). Table 2 shows a synthesis of these results.

The mean duration of the HIIT, SIT and MOD protocols included in the analyses of total body fat percentage (%) were 28 min, 18 min and 38 min, respectively. The percentage

reductions of total body fat percentage (%) for these protocols were, on average, 4.6%, 3.5% and 3.5%, respectively. On average, HIIT, SIT and MOD protocols included in the analysis on total absolute fat mass (kg) lasted 25 min, 23 min and 41 min, respectively. The percentage reductions of total absolute fat mass (kg) were, on average, 6.0%, 6.2% and 3.4%, respectively.

Sensitivity analyses and publication bias

A sensitivity analysis showed that a significant effect ($p<0.05$) of HIIT/SIT on total absolute fat mass (kg) remained after removal of each one of the included studies, with evidence of significant heterogeneity ($p<0.05$). Funnel plots and Begg's tests for all analyses determined no indication of publication bias.

DISCUSSION

The present study analysed data from studies that compared the effects of interval training and MOD on body adiposity in humans. The analysis combined 41 studies (36 for meta-analysis) involving a total of 1115 participants. Most studies included in the meta-analysis (86.1%) involved a small sample size (<20 participants per intervention); therefore, the lack of statistical power might have prevented the detection of between-group differences in isolated studies. Notwithstanding, by pooling the data, we did not find superiority of either interval training or MOD in the reduction of total body fat percentage (%), as previously reported in an earlier meta-analysis.²¹ However, when compared with MOD, we found a superiority of interval training in the reduction of absolute total fat mass (kg). Indeed, both interval training and MOD were similarly beneficial in eliciting small improvements in total body fat percentage (%) (HIIT/SIT: −1.50%; MOD: −1.44%) and in total absolute fat mass (kg) (HIIT/SIT: −1.58 kg; MOD: −1.13 kg). However, a significant difference was found between SIT and MOD in total absolute fat mass (kg) (online supplementary figure S3B).

As a result of the sensitivity analysis that removed each study one by one, we noted that the significant difference favouring interval training for total absolute fat mass (kg) reduction remained. To better understand the factors that might influence the results, we critically reviewed individual studies that favoured interval training or MOD. It is noteworthy that the studies were selected based on their impact on our meta-analysis and not necessarily the results reported in the article.

As for the data that supported MOD for a greater reduction in total body fat percentage (%), the study by Buchan *et al*^{61 62} involved adolescents and started with four 30 s 'all-out' running bouts interspaced by 30 s of rest, progressing to six bouts with 20 s of rest. This protocol, however, seems unfeasible, since the recommended recovery between bouts in similar protocols is ~8 times the duration of the effort, such that 30 s maximum efforts are usually followed by 4 min of rest.²⁰ As such, it seems unlikely the participants in the HIIT group in the study by Buchan *et al*^{61 62} were able to maintain maximal effort across the exercise bouts. Koubaa *et al*⁶⁶ reported using running intervals of 2 min at 80% of $\dot{V}O_{2\max}$ followed by 1 min of rest for interval training in adolescents. However, they do not report information about the number of bouts nor about rest intervals, which makes it difficult to analyse the protocol. Moreover, neither Buchan *et al*^{61 62} nor Koubaa *et al*⁶⁶ provide data on dietary and physical activity control.

Another study in favour of MOD is by Nybo *et al*,⁵⁹ which involved untrained men. During interval training, participants were instructed to exceed 95% of the HRmax at the end of 2 min of running and then rest for 2 min. However, considering that

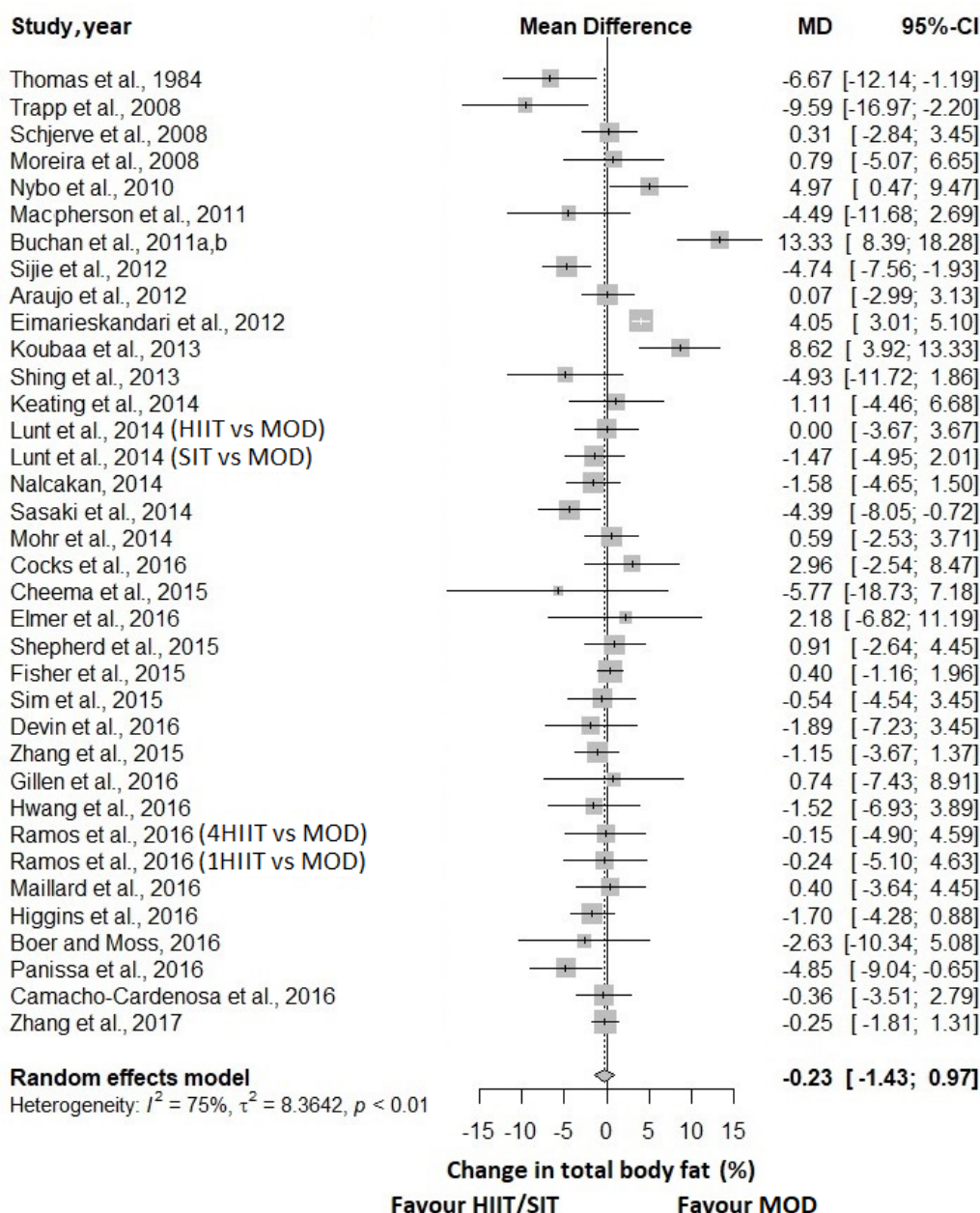


Figure 2 Main effects of HIIT/SIT versus MOD on total body fat (%). 4HIIT, high-intensity interval training (four bouts); HIIT, high-intensity interval training; MD, mean of differences; MOD, moderate-intensity interval training; SIT, sprint interval training.

the intensity of effort was controlled only at the end of each bout, it is not possible to be certain about the intensity of effort maintained during each interval. If we consider that HR progressively increases at a constant rate, it might be possible that the participants spent most of the time at an intensity of effort lower than recommended. Moreover, although the participants were oriented to maintain their habitual lifestyle and dietary practices, the authors did not control for this variable.

Some studies favoured interval training for per cent body fat loss. Panissa *et al.*⁹⁰ used a 22 min protocol with 1 min at 90% and 30 s at 60% of HRmax. However, it seems again unfeasible for participants to achieve the prescribed intensity of effort based on the percentage of HRmax since both the times taken to increase and decrease HR seem too short. For example, in the study of Ramos *et al.*,⁸⁶ participants took 2 min to reach a similar intensity of effort, and a previous study showed that HR decreases

~30 beats per minute in the first minute after intense exercise.⁹⁴ The results of the study by Thomas *et al.*⁵⁴ also favoured interval training for reductions in per cent body fat in a mixed sample of men and women. MOD involved running for 3.2 or 6.4 km at 75% of HRmax, while interval training involved eight bouts of running for 1 min at 90% HRmax followed by 3 min of walking. However, a limitation of this study was the absence of diet and physical activity control. Macpherson *et al.*⁶⁰ compared the effects of SIT (4–6 'all-out' efforts of 30 s in a manually driven treadmill with 4 min of rest) and MOD (30–60 min running at 60% $\text{VO}_{2\text{peak}}$) in a mixed sample of physically active men and women. Their data pointed towards greater decreases in per cent body fat for SIT; however, while the authors reported to have encouraged the participants to maintain their physical activity and diet patterns, there were no objective measures of these variables.

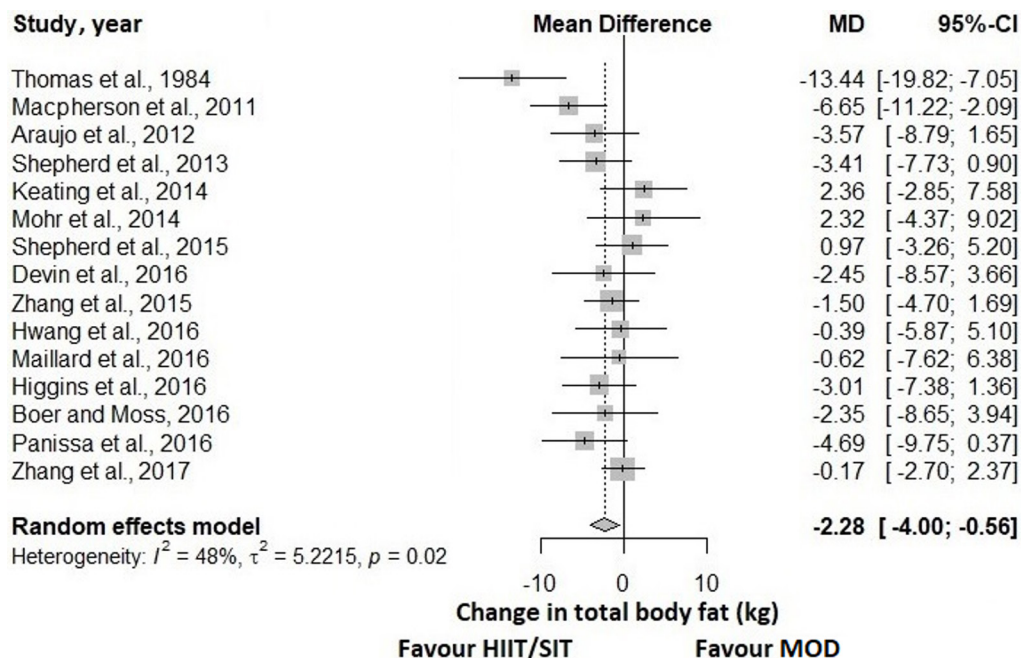


Figure 3 Main effects of HIIT/SIT versus MOD on total absolute fat mass (kg). HIIT, high-intensity interval training; MD, mean of differences; MOD, moderate-intensity interval training; SIT, sprint interval training.

These individual studies highlight the difficulty of drawing general conclusions about the application and effects of interval training on body composition. The inconsistent results might be linked to factors such as habitual diet and physical activity behaviours, since only 23 (~56%) and 19 (~46%) of the included studies reported diet and physical activity control, respectively. Another aspect that needs to be considered is the quality of studies performing interval training and MOD. When considering only the included studies with middle quality, intervention duration less than 12 weeks or with participants' age less than 30 years, our results found a significant reduction in absolute total fat mass (kg) favouring interval training, although no significant difference was found on total body fat percentage (%). This suggests an influence of the methodological quality of the studies and participants' characteristics on the results. Moreover, other aspects that might influence weight loss, such as hormonal status,⁹⁵ sleeping patterns⁹⁶ and mood disorders,⁹⁷ are not usually analysed in these studies.

With regard to the factors inherent to interval training, the absence of adequate control for supervision, intensity of effort and the effort to rest ratio might be associated with at least some of the inconsistent results. The subgroup meta-analysis demonstrated that improvements in total absolute fat mass (kg) caused by interval training are higher with supervision during interval training protocols, providing evidence that supervision during interval training is an important variable for total absolute fat mass (kg) reduction. This might occur because it can help interval training practitioners to train with higher intensity of effort.⁹⁸ Considering that supervision might guarantee adherence to the prescribed protocol, the results provided by studies with supervised sessions are probably more reliable. Therefore, it is important that interval training studies consider providing supervision to guarantee accountability.

In this sense, some examples of the inconsistency with regard to the intensity of effort can be obtained from the analysis of individual studies. For example, Keating *et al.*⁷⁰ reported that their protocol was based on the study by Little *et al.*⁹⁹ which

used 10 bouts of 60 s at a load that elicited 90% of HRmax interspaced by 60 s of recovery. However, in the study by Keating *et al.*⁷⁰ HIIT was performed at 120% of iVO_{2max} with 30–60 s duration and 120–180 s of rest. On the other hand, at 120% of iVO_{2max} , a previous study used seven bouts of 30 s interspaced with 15 s of rest.¹³

In addition, another interesting example of the possible influence of the control of intensity of effort on the results might be found in Ramos *et al.*⁸⁶ While the protocol was reported to involve four bouts of 4 min at 85%–95% of HRpeak with 3 min intervals, the participants took 2 min to reach the targeted intensity. Therefore, the protocol seems to have involved 2 min of the actual prescribed intensity of effort. In other words, the time at target intensity of effort was not reached as planned, resulting in a lower effort to rest ratio than intended. We would like to note that these individual observations do not invalidate or question the merit of previous studies. These are only some aspects that might explain the large inconsistency among the results of interval training and which we must consider when analysing and reproducing previous studies.

Our results found that the effect of interval training protocols when performed using walking/jogging/running modalities on total absolute fat mass (kg) is greater than for MOD with the same modalities. However, the number ($n=5$) of studies included in this analysis was low,^{54 60 64 69 82} and only the study by Zhang *et al.*⁸² monitored and reported to control diet and habitual activity. Moreover, our data also showed an influence of exercise supervision, and a separate analysis showed that interval training resulted in greater loss of total absolute fat mass (kg) than MOD when training was supervised. Possibly, supervision might influence accountability, influencing adherence to the prescribed intensity of effort.⁹⁸ Indeed, for other exercise modalities, such as resistance training, supervision has been shown to impact significantly on the intensity of effort and outcomes.⁹⁸

Separate analyses with HIIT and SIT showed that 'all-out' SIT promotes greater total absolute fat mass (kg) reduction than MOD. The greater decreases in fat loss promoted by SIT might

Table 2 Summary of HIIT/SIT versus MOD subgroup meta-analysis on body composition

Outcome (subgroup)	Between-group effects			Heterogeneity	
	Studies (n)	MD (95% CI)	P value	I ² (%)	P value
Total body fat (%)					
Mode: HIIT	23	−0.35 (−1.90 to 1.21)	0.66	78	<0.01
Mode: SIT	13	−0.04 (−2.02 to 1.94)	0.97	71	<0.01
Modality: walking/jogging/running	14	−1.03 (−1.33 to 3.38)	0.39	86	<0.01
Modality: cycling	16	−0.88 (−1.93 to 0.17)	0.10	26	0.16
Sex: male	10	0.86 (−1.36 to 3.07)	0.45	65	<0.01
Sex: female	9	−1.36 (−3.85 to 1.14)	0.29	89	<0.01
Age <30 years	19	−0.23 (−2.08 to 1.62)	0.81	86	<0.01
Age ≥30 years	16	0.15 (−0.94 to 1.23)	0.79	0	0.92
BMI <30 kg/m ²	18	−0.81 (−2.54 to 0.93)	0.36	70	<0.01
BMI ≥30 kg/m ²	10	1.15 (−0.98 to 3.28)	0.29	77	<0.01
Study quality: low	12	0.44 (−2.11 to 2.98)	0.74	85	<0.01
Study quality: middle	16	−0.52 (−2.29 to 1.26)	0.57	67	<0.01
Study quality: high	8	−0.89 (−2.13 to 0.36)	0.16	0	0.96
Intervention duration <12 months	15	−0.03 (−2.11 to 2.05)	0.98	84	<0.01
Intervention duration ≥12 months	21	−0.38 (−1.66 to 0.91)	0.56	53	<0.01
Supervision: yes	26	−0.71 (−1.85 to 0.44)	0.23	59	<0.01
Supervision: no	10	1.06 (−1.60 to 3.72)	0.43	80	<0.01
Total fat mass (kg)					
Mode: HIIT	10	−1.96 (−4.19 to 0.26)	0.08	56	0.02
Mode: SIT	5	−3.22 (−5.71 to −0.73)	0.01	18	0.30
Modality: walking/jogging/running	5	−5.18 (−8.73 to −1.63)	<0.01	67	0.02
Modality: cycling	9	−1.17 (−2.64 to 0.29)	0.12	0	0.51
Sex: male	2	−0.14 (−3.62 to 3.34)	0.94	0	0.37
Sex: female	6	−1.23 (−2.83 to 0.38)	0.14	0	0.51
Age <30 years	8	−3.92 (−6.36 to −1.49)	<0.01	63	<0.01
Age ≥30 years	7	0.19 (−1.94 to 2.32)	0.86	0	0.85
BMI <30 kg/m ²	8	−1.36 (−3.03 to 0.26)	0.10	0	0.54
BMI ≥30 kg/m ²	3	−2.75 (−5.77 to 0.27)	0.08	0	0.79
Study quality: low	2	−5.59 (−21.03 to 9.85)	0.48	91	<0.01
Study quality: middle	10	−2.03 (−3.60 to −0.47)	0.01	12	0.33
Study quality: high	3	−1.10 (−3.73 to 1.54)	0.42	20	0.29
Intervention duration <12 months	7	−2.82 (−4.79 to −0.84)	<0.01	16	0.31
Intervention duration ≥12 months	8	−1.88 (−4.67 to 0.90)	0.19	62	<0.01
Supervision: yes	14	−2.21 (−4.03 to 0.38)	0.02	51	0.01
Supervision: no	1	–	–	–	–

Significant p values are indicated in bold.

BMI, body mass index; HIIT, high-intensity interval training; MD, mean of differences; MOD, moderate-intensity continuous training; SIT, sprint interval training.

be due to the increases in postexercise fat oxidation, which seems to be associated with glycogen depletion.^{25 100} Indeed, vigorous exercise may be mediated by a more pronounced increase in the skeletal muscle oxidative capacity and by a sympathoadrenal stimulation.¹⁰¹ Thus, protocols that rely more on the glycolytic system might be more beneficial to body fat reductions.^{102 103}

In general, although our findings suggest that MOD provides similar benefits to interval training for total body fat percentage (%) reduction, interval training might be an efficacious, ‘time-efficient’ exercise strategy for body fat management, since the MOD protocols examined in the included studies usually had a greater duration than interval training protocols and provide similar reductions in total body fat percentage (%). For example, MOD protocols lasted on average 38 min (average of 35 sessions × 38 min/session = 1330 min) and provided a reduction of 3.5% in total body fat percentage (%), while HIIT protocols lasted on average 28 min (average

of 33 sessions × 28 min/session = 924 min) and provided a reduction of 4.6%, and SIT protocols lasted on average 18 min (average of 29 sessions × 18 min/session = 526 min) and provided a reduction of 3.5% in total body fat percentage (%). In other words, MOD protocols provided a reduction of ‘0.0026% per minute’, while HIIT and SIT protocols provided a reduction of ‘0.0050% and 0.0067% per minute’ in total body fat percentage (%), respectively. The analysis showed that interval training promotes greater reductions in total absolute fat mass (kg) than MOD, despite requiring less time to be performed. However, it is important to be aware of the possible risks and caveats associated with higher intensity training. For example, it might increase the risk of injury and impose higher cardiovascular stress. Adherence should also be examined, as higher intensity protocols can result in higher discomfort.

Review

A common criticism of meta-analysis is the combination of largely heterogeneous studies that have important methodological differences, which can influence the reported effects particularly when the number of included studies is low.^{104 105} According to Grindem *et al*,¹⁰⁶ heterogeneity is a key factor in the decision to pool or not to pool the results of available studies, which makes it a challenging issue in systematic reviews. The included studies presented relatively high heterogeneity in the total meta-analysis of total body fat percentage (%) and total absolute fat mass (kg), and this may be a reflection of the large heterogeneity in exercise protocols used in the included studies. Broadly speaking, the different protocols (HIIT/SIT or MOD) seem similarly effective in modulating body adiposity in humans; however, the varied approaches used make it difficult to draw general conclusions and recommendations about the 'ideal' interval training or MOD protocol. Therefore, clinicians must be careful when interpreting these results and applying them to their practice. Future studies must improve their methodological quality, sample size and method of assessment of change in total body fat to provide more compelling evidence in favour of a specific protocol, or to elucidate the principles of protocol design that appear to have the greatest influence on outcomes.

CONCLUSION

The present systematic review with meta-analysis showed that interval training provides benefits similar to MOD in total body fat percentage (%) reduction; however, interval training provided a greater total absolute fat mass (kg) reduction than MOD. SIT resulted in greater total absolute fat loss when compared with MOD. A number of factors may positively influence the effects of interval training on total absolute fat mass, including supervision of exercise, walking/running/jogging as the exercise of

choice, age (<30 years), study quality and intervention duration (<12 weeks). In general, our findings suggest that the 'signal in the noise' is the similar effects of interval training and MOD on total body fat percentage (%) management and the superiority of interval training for total absolute fat mass (kg) reduction, yet that these effects can be produced in a 'time-efficient' manner when using interval training.

Contributors RBV and JPAN carried out the screenings and reviews. RBV and VSC carried out the analysis of the articles. RBV and PG drafted and revised the manuscript. CABdL, VSC, JS, JPF and PG revised the manuscript. All authors read and approved the final manuscript.

Funding This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Competing interests None declared.

Patient consent Not required.

Provenance and peer review Not commissioned; externally peer reviewed.

REFERENCES

- Shaw KA, Gennat HC, O'Rourke P, *et al*. Exercise for overweight or obesity. *Cochrane Database Syst Rev* 2006;73.
- Franz MJ, VanWormer JJ, Crain AL, *et al*. Weight-loss outcomes: a systematic review and meta-analysis of weight-loss clinical trials with a minimum 1-year follow-up. *J Am Diet Assoc* 2007;107:1755–67.
- Donnelly JE, Blair SN, Jakicic JM, *et al*. Appropriate physical activity intervention strategies for weight loss and prevention of weight regain for adults. *Med Sci Sports Exerc* 2009;41:459–71.
- Saris WHM, Blair SN, van Baak MA, *et al*. How much physical activity is enough to prevent unhealthy weight gain? Outcome of the IASO 1st Stock Conference and consensus statement. *Obesity Reviews* 2003;4:101–14.
- Trumbo P, Schlicker S, Yates AA, *et al*. Dietary reference intakes for energy, carbohydrate, fiber, fat, fatty acids, cholesterol, protein and amino acids. *J Am Diet Assoc* 2002;102:1621–30.
- Vella CA, Taylor K, Drummer D. High-intensity interval and moderate-intensity continuous training elicit similar enjoyment and adherence levels in overweight and obese adults. *Eur J Sport Sci* 2017;17:1203–11.
- Weston KS, Wisloff U, Coombes JS. High-intensity interval training in patients with lifestyle-induced cardiometabolic disease: a systematic review and meta-analysis. *Br J Sports Med* 2014;48:1227–34.
- Gibala MJ, Little JP, MacDonald MJ, *et al*. Physiological adaptations to low-volume, high-intensity interval training in health and disease. *J Physiol* 2012;590:1077–84.
- Gibala MJ. High-intensity interval training: a time-efficient strategy for health promotion? *Curr Sports Med Rep* 2007;6:211–3.
- Norton K, Norton L, Sadgrove D. Position statement on physical activity and exercise intensity terminology. *J Sci Med Sport* 2010;13:496–502.
- Fox EL, Bartels RL, Billings CE, *et al*. Intensity and distance of interval training programs and changes in aerobic power. *Med Sci Sports* 1973;5:18–22.
- MacInnis MJ, Gibala MJ. Physiological adaptations to interval training and the role of exercise intensity. *J Physiol* 2017;595:2915–30.
- Matsuo T, Saotome K, Seino S, *et al*. Effects of a low-volume aerobic-type interval exercise on VO₂max and cardiac mass. *Med Sci Sports Exerc* 2014;46:42–50.
- Guiraud T, Nigam A, Gremeaux V, *et al*. High-intensity interval training in cardiac rehabilitation. *Sports Medicine* 2012;42:587–605.
- Jelleyman C, Yates T, O'Donovan G, *et al*. The effects of high-intensity interval training on glucose regulation and insulin resistance: a meta-analysis. *Obesity Reviews* 2015;16:942–61.
- Molmen-Hansen HE, Stolen T, Tjonna AE, *et al*. Aerobic interval training reduces blood pressure and improves myocardial function in hypertensive patients. *Eur J Prev Cardiol* 2012;19:151–60.
- Kilpatrick MW, Greeley SJ, Collins LH. The impact of continuous and interval cycle exercise on affect and enjoyment. *Res Q Exerc Sport* 2015;86:244–51.
- Jung ME, Bourne JE, Beauchamp MR, *et al*. High-intensity interval training as an efficacious alternative to moderate-intensity continuous training for adults with prediabetes. *J Diabetes Res* 2015;2015:1–9.
- Burgomaster KA, Howarth KR, Phillips SM, *et al*. Similar metabolic adaptations during exercise after low volume sprint interval and traditional endurance training in humans. *J Physiol* 2008;586:151–60.
- Gibala MJ, Little JP, Van Essen M, *et al*. Short-term sprint interval versus traditional endurance training: similar initial adaptations in human skeletal muscle and exercise performance. *J Physiol* 2006;575:901–11.
- Keating SE, Johnson NA, Mielke GI, *et al*. A systematic review and meta-analysis of interval training versus moderate-intensity continuous training on body adiposity. *Obesity Reviews* 2017;18:943–64.

What is already known

- Physical activity may be a useful tool to reduce body adiposity; however, many people fail to adhere to exercise programmes due to lack of time or lack of results.
- Interval training is an attractive alternative to address overweight and obesity given its potential to offer benefits similar to moderate-intensity continuous training while requiring less time.
- There are currently many different approaches to performing interval training, yet there is no consensus as to which training method or protocol is 'best' for reducing body adiposity.

What are the new findings

- Interval training and moderate-intensity continuous training provide similar benefits for body fat percentage reduction; however, interval training provides greater reductions in total absolute fat mass.
- Supervision, walking/running/jogging, age, study quality and intervention duration seem to favourably influence the decreases in body adiposity observed from interval training programmes.
- There is great methodological diversity among interval training protocols in the literature, which makes it difficult to generally recommend that one particular protocol is 'best' for modulating body adiposity in humans.

- 22 Boutcher SH. High-intensity intermittent exercise and fat loss. *J Obes* 2011;2011:1–10.
- 23 Tremblay A, Simoneau J-A, Bouchard C. Impact of exercise intensity on body fatness and skeletal muscle metabolism. *Metabolism* 1994;43:814–8.
- 24 Trapp EG, Chisholm DJ, Freund J, et al. The effects of high-intensity intermittent exercise training on fat loss and fasting insulin levels of young women. *Int J Obes* 2008;32:684–91.
- 25 Kiens B, Richter EA. Utilization of skeletal muscle triacylglycerol during postexercise recovery in humans. *Am J Physiol Endocrinol Metab* 1998;275:E332–E337.
- 26 Kelly B, King JA, Goerlach J, et al. The impact of high-intensity intermittent exercise on resting metabolic rate in healthy males. *Eur J Appl Physiol* 2013;113:3039–47.
- 27 Viana RB, de Lira CAB, Naves JPA, et al. Can we draw general conclusions from interval training studies? *Sports Med* 2018;48:2001–9.
- 28 Moher D, Liberati A, Tetzlaff J, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *BMJ* 2009;339:b2535.
- 29 Booth A, Clarke M, Dooley G, et al. The nuts and bolts of PROSPERO: an international prospective register of systematic reviews. *Syst Rev* 2012;1:1–8.
- 30 Downs SH, Black N. The feasibility of creating a checklist for the assessment of the methodological quality both of randomised and non-randomised studies of health care interventions. *J Epidemiol Community Health* 1998;52:377–84.
- 31 Higgins J, Green S. *Cochrane handbook for systematic reviews of interventions*. 5. 1: The Cochrane Collaboration, 2008. <http://training.cochrane.org/handbook>.
- 32 Terada T, Friesen A, Chahal BS, et al. Feasibility and preliminary efficacy of high intensity interval training in type 2 diabetes. *Diabetes Res Clin Pract* 2013;99:120–9.
- 33 Ruffino JS, Songson P, Haggett M, et al. A comparison of the health benefits of reduced-exertion high-intensity interval training (REHIT) and moderate-intensity walking in type 2 diabetes patients. *Appl Physiol Nutr Metab* 2017;42:202–8.
- 34 Boer P-H, Meeus M, Terblanche E, et al. The influence of sprint interval training on body composition, physical and metabolic fitness in adolescents and young adults with intellectual disability: a randomized controlled trial. *Clin Rehabil* 2014;28:221–31.
- 35 García MV, Ibáñez AV, Poveda DC, et al. Efecto de 12 sesiones de un entrenamiento interválico de alta intensidad sobre la composición corporal en adultos jóvenes. *Nutr Hosp* 2016;33:637–43.
- 36 Mazurek K, Zmijewski P, Krawczyk K, et al. High intensity interval and moderate continuous cycle training in a physical education programme improves health-related fitness in young females. *Biol Sport* 2016;33:139–44.
- 37 Giannaki CD, Aghamias G, Sakakis P, et al. Eight weeks of a combination of high intensity interval training and conventional training reduce visceral adiposity and improve physical fitness : a group-based intervention. *J Sports Med Phys Fitness* 2015;56:483–90.
- 38 Rowan CP, Riddell MC, Gledhill N, et al. Aerobic exercise training modalities and prediabetes risk reduction. *Med Sci Sports Exerc* 2017;49:403–12.
- 39 Thøgersen-Ntoumani C, Shepherd SO, Ntoumanis N, et al. Intrinsic motivation in two exercise interventions: Associations with fitness and body composition. *Health Psychology* 2016;35:195–8.
- 40 Matsuo T, So R, Shimojo N, et al. Effect of aerobic exercise training followed by a low-calorie diet on metabolic syndrome risk factors in men. *Nutr Metab Cardiovasc Dis* 2015;25:832–8.
- 41 Kong Z, Sun S, Liu M, et al. Short-term high-intensity interval training on body composition and blood glucose in overweight and obese young women. *J Diabetes Res* 2016;2016:1–9.
- 42 Kong Z, Fan X, Sun S, et al. Comparison of high-intensity interval training and moderate-to-vigorous continuous training for cardiometabolic health and exercise enjoyment in obese young women: A randomized controlled trial. *PLoS One* 2016;11:e0158589.
- 43 Cooper JHF, Collins BEG, Adams DR, et al. Limited effects of endurance or interval training on visceral adipose tissue and systemic inflammation in sedentary middle-aged men. *J Obes* 2016;2016:1–10.
- 44 Støa EM, Meling S, Nyhus L-K, et al. High-intensity aerobic interval training improves aerobic fitness and HbA1c among persons diagnosed with type 2 diabetes. *Eur J Appl Physiol* 2017;117:455–67.
- 45 Venables MC, Jeukendrup AE. Endurance training and obesity: Effect on substrate metabolism and insulin sensitivity. *Med Sci Sports Exerc* 2008;40:495–502.
- 46 Campbell L, Wallman K, Green D. The effects of intermittent exercise on physiological outcomes in an obese population: Continuous versus interval walking. *J Sport Sci Med* 2010;9:24–30.
- 47 Karstoft K, Winding K, Knudsen SH, et al. The effects of free-living interval-walking training on glycemic control, body composition, and physical fitness in type 2 diabetic patients: a randomized, controlled trial. *Diabetes Care* 2013;36:228–36.
- 48 Aksoy S, Findikoglu G, Ardic F, et al. Effect of 10-week supervised moderate-intensity intermittent vs. continuous aerobic exercise programs on vascular adhesion molecules in patients with heart failure. *Am J Phys Med Rehabil* 2015;94:898–911.
- 49 Grossman JAC, Payne EK. A randomized comparison study regarding the impact of short-duration, high-intensity exercise and traditional exercise on anthropometric and body composition measurement changes in post-menopausal women – A pilot study. *Post Reprod Health* 2016;22:14–19.
- 50 Tjønnå AE, Lee SJ, Rognmo Ø, et al. Aerobic interval training versus continuous moderate exercise as a treatment for the metabolic syndrome: a pilot study. *Circulation* 2008;118:346–54.
- 51 Connolly LJ, Nordsborg NB, Nyberg M, et al. Low-volume high-intensity swim training is superior to high-volume low-intensity training in relation to insulin sensitivity and glucose control in inactive middle-aged women. *Eur J Appl Physiol* 2016;116:1889–97.
- 52 Kim HK, Hwang CL, Yoo JK, et al. All-extremity exercise training improves arterial stiffness in older adults. *Med Sci Sports Exerc* 2017;49:1404–11.
- 53 Di Blasio A, Izzicupo P, D'Angelo E, et al. Effects of patterns of walking training on metabolic health of untrained postmenopausal women. *J Aging Phys Act* 2014;22:482–9.
- 54 Thomas TR, Adeniran SB, Etheridge GL. Effects of different running programs on VO2max, percent fat, and plasma lipids. *Can J Appl Sport Sci* 1984;9:55–62.
- 55 Mäder U, Roth P, Furrer R, et al. Influence of continuous and discontinuous training protocols on subcutaneous adipose tissue and plasma substrates. *Int J Sports Med* 2001;22:344–9.
- 56 Schjerve IE, Tyldum GA, Tjønnå AE, et al. Both aerobic endurance and strength training programmes improve cardiovascular health in obese adults. *Clin Sci* 2008;115:283–93.
- 57 Moreira MM, Souza HPCde, Schwingel PA, et al. Efeitos do exercício aeróbico e anaeróbico em variáveis de risco cardíaco em adultos com sobrepeso. *Arq Bras Cardiol* 2008;91:219–26.
- 58 Wallman K, Plant LA, Rakimov B, et al. The effects of two modes of exercise on aerobic fitness and fat mass in an overweight population. *Res Sports Med* 2009;17:156–70.
- 59 Nybo L, Sundstrup E, Jakobsen MD, et al. High-intensity training versus traditional exercise interventions for promoting health. *Med Sci Sports Exerc* 2010;42:1951–8.
- 60 Macpherson RE, Hazell TJ, Olver TD, et al. Run sprint interval training improves aerobic performance but not maximal cardiac output. *Med Sci Sports Exerc* 2011;43:115–22.
- 61 Buchan DS, Ollis S, Young JD, et al. The effects of time and intensity of exercise on novel and established markers of CVD in adolescent youth. *Am J Human Biol* 2011;23:517–26.
- 62 Buchan DS, Ollis S, Thomas NE, et al. Physical activity interventions: effects of duration and intensity. *Scand J Med Sci Sports* 2011;21:e341–e350.
- 63 Sijie T, Hainai Y, Fengying Y, et al. High intensity interval exercise training in overweight young women. *J Sports Med Phys Fitness* 2012;52:255–62.
- 64 Corte de Araujo AC, Roschel H, Picanço AR, et al. Similar health benefits of endurance and high-intensity interval training in obese children. *PLoS One* 2012;7:e42747.
- 65 Eimariesskandari R, Zilaebouri S, Zilaebouri M, et al. Comparing two modes of exercise training with different intensity on body composition in obese young girls. *Ovidius Univ Ann Ser Phys Educ Sport Mov Heal* 2012;12:473–8.
- 66 Koubaa A, et al. Effect of intermittent and continuous training on body composition cardiorespiratory fitness and lipid profile in obese adolescents. *Isr Journal Of Pharmacy* 2013;3:31–7.
- 67 Earnest CP, Lupo M, Thibodaux J, et al. Interval training in men at risk for insulin resistance. *Int J Sports Med* 2013;34:355–63.
- 68 Shing CM, Webb JJ, Driller MW, et al. Circulating adiponectin concentration and body composition are altered in response to high-intensity interval training. *J Strength Cond Res* 2013;27:2213–8.
- 69 Shepherd SO, Cocks M, Tipton KD, et al. Sprint interval and traditional endurance training increase net intramuscular triglyceride breakdown and expression of perilipin 2 and 5. *J Physiol* 2013;591:657–75.
- 70 Keating SE, Machan EA, O'Connor HT, et al. Continuous exercise but not high intensity interval training improves fat distribution in overweight adults. *J Obes* 2014;2014:1–12.
- 71 Lunt H, Draper N, Marshall HC, et al. High intensity interval training in a real world setting: a randomized controlled feasibility study in overweight inactive adults, measuring change in maximal oxygen uptake. *PLoS One* 2014;9:e83256.
- 72 Nalcakan GR. The effects of sprint interval vs. Continuous endurance training on physiological and metabolic adaptations in young healthy adults. *J Hum Kinet* 2014;44:97–109.
- 73 Sasaki H, Morishima T, Hasegawa Y, et al. 4 weeks of high-intensity interval training does not alter the exercise-induced growth hormone response in sedentary men. *Springerplus* 2014;3:336–9.
- 74 Mohr M, Nordsborg NB, Lindenskov A, et al. High-intensity intermittent swimming improves cardiovascular health status for women with mild hypertension. *Biomed Res Int* 2014;2014:1–9.
- 75 Cocks M, Shaw CS, Shepherd SO, et al. Sprint interval and moderate-intensity continuous training have equal benefits on aerobic capacity, insulin sensitivity, muscle capillarisation and endothelial eNOS/NAD(P)H oxidase protein ratio in obese men. *J Physiol* 2016;594:2307–21.
- 76 Cheema BS, Davies TB, Stewart M, et al. The feasibility and effectiveness of high-intensity boxing training versus moderate-intensity brisk walking in adults with abdominal obesity: a pilot study. *BMC Sports Sci Med Rehabil* 2015;7:3.

- 77 Elmer DJ, Laird RH, Barberio MD, *et al.* Inflammatory, lipid, and body composition responses to interval training or moderate aerobic training. *Eur J Appl Physiol* 2016;116:601–9.
- 78 Shepherd SO, Wilson OJ, Taylor AS, *et al.* Low-volume high-intensity interval training in a gym setting improves cardio-metabolic and psychological health. *PLoS One* 2015;10:e0139056.
- 79 Fisher G, Brown AW, Bohan Brown MM, *et al.* High intensity interval- vs moderate intensity- training for improving cardiometabolic health in overweight or obese males: A randomized controlled trial. *PLoS One* 2015;10:e0138853.
- 80 Sim AY, Wallman KE, Fairchild TJ, *et al.* Effects of High-Intensity Intermittent Exercise Training on Appetite Regulation. *Med Sci Sports Exerc* 2015;47:2441–9.
- 81 Devin JL, Sax AT, Hughes GI, *et al.* The influence of high-intensity compared with moderate-intensity exercise training on cardiorespiratory fitness and body composition in colorectal cancer survivors: a randomised controlled trial. *J Cancer Surviv* 2016;10:467–79.
- 82 Zhang H, Tong T K, Qiu W, *et al.* Effect of high-intensity interval training protocol on abdominal fat reduction in overweight Chinese women: a randomized controlled trial. *Kineziologija* 2015;47:57–66.
- 83 Gillen JB, Martin BJ, MacInnis MJ, *et al.* Twelve weeks of sprint interval training improves indices of cardiometabolic health similar to traditional endurance training despite a five-fold lower exercise volume and time commitment. *PLoS One* 2016;11:e0154075–14.
- 84 Martins C, Kazakova I, Ludviksen M, *et al.* High-intensity interval training and isocaloric moderate-intensity continuous training result in similar improvements in body composition and fitness in obese individuals. *Int J Sport Nutr Exerc Metab* 2016;26:197–204.
- 85 Hwang C-L, Yoo J-K, Kim H-K, *et al.* Novel all-extremity high-intensity interval training improves aerobic fitness, cardiac function and insulin resistance in healthy older adults. *Exp Gerontol* 2016;82:112–9.
- 86 Ramos JS, Dalleck LC, Borrani F, *et al.* The effect of different volumes of high-intensity interval training on proinsulin in participants with the metabolic syndrome: a randomised trial. *Diabetologia* 2016;59:2308–20.
- 87 Maillard F, Rousset S, Pereira B, *et al.* High-intensity interval training reduces abdominal fat mass in postmenopausal women with type 2 diabetes. *Diabetes Metab* 2016;42:433–41.
- 88 Higgins S, Fedewa MV, Hathaway ED, *et al.* Sprint interval and moderate-intensity cycling training differentially affect adiposity and aerobic capacity in overweight young-adult women. *Appl Physiol Nutr Metab* 2016;41:1177–83.
- 89 Boer PH, Moss SJ. Effect of continuous aerobic vs. interval training on selected anthropometrical, physiological and functional parameters of adults with Down syndrome. *J Intellect Disabil Res* 2016;60:322–34.
- 90 Panissa VL, Alves ED, Salerno GP, *et al.* Can short-term high-intensity intermittent training reduce adiposity? *Sport Sci Health* 2016;12:99–104.
- 91 Camacho-Cardenosa A, Brazo-Sayavera J, Camacho-Cardenosa M, *et al.* Effects of high intensity interval training on fat mass parameters in adolescents. *Rev Esp Salud Publica* 2016;90:1–9.
- 92 Galedari M, Azarbayjani MA, Peeri M. Effects of type of exercise along with caloric restriction on plasma apelin 36 and HOMA-IR in overweight men. *Sci Sports* 2017;32:e137–e145.
- 93 Zhang H, Tong TK, Qiu W, *et al.* Comparable effects of high-intensity interval training and prolonged continuous exercise training on abdominal visceral fat reduction in obese young women. *J Diabetes Res* 2017;2017:1–9.
- 94 Antelmi I, Chuang EY, Grupi CJ, *et al.* Heart rate recovery after treadmill electrocardiographic exercise stress test and 24-hour heart rate variability in healthy individuals. *Arq Bras Cardiol* 2008;90:380–5.
- 95 Armellini F, Zamboni M, Bosello O. Hormones and body composition in humans: clinical studies. *Int J Obes* 2000;24(S2):S18–S21 <http://www.ncbi.nlm.nih.gov/pubmed/10997601>.
- 96 Shlisky JD, Hartman TJ, Kris-Etherton PM, *et al.* Partial sleep deprivation and energy balance in adults: an emerging issue for consideration by dietetics practitioners. *J Acad Nutr Diet* 2012;112:1785–97.
- 97 Bond DJ, Kunz M, Torres IJ, *et al.* The association of weight gain with mood symptoms and functional outcomes following a first manic episode: prospective 12-month data from the Systematic Treatment Optimization Program for Early Mania (STOP-EM). *Bipolar Disord* 2010;12:616–26.
- 98 Gentil P, Bottaro M. Influence of supervision ratio on muscle adaptations to resistance training in nontrained subjects. *J Strength Cond Res* 2010;24:639–43.
- 99 Little JP, Gillen JB, Percival ME, *et al.* Low-volume high-intensity interval training reduces hyperglycemia and increases muscle mitochondrial capacity in patients with type 2 diabetes. *J Appl Physiol* 2011;111:1554–60.
- 100 Withers RT, Sherman WM, Clark DG, *et al.* Muscle metabolism during 30, 60 and 90 s of maximal cycling on an air-braked ergometer. *Eur J Appl Physiol Occup Physiol* 1991;63:354–62.
- 101 Yoshioka M, Doucet E, St-Pierre S, *et al.* Impact of high-intensity exercise on energy expenditure, lipid oxidation and body fatness. *Int J Obes Relat Metab Disord* 2001;25:332–9.
- 102 Tucker WJ, Angadi SS, Gaesser GA. Excess postexercise oxygen consumption after high-intensity and sprint interval exercise, and continuous steady-state exercise. *J Strength Cond Res* 2016;30:3090–7.
- 103 Whyte LJ, Ferguson C, Wilson J, *et al.* Effects of single bout of very high-intensity exercise on metabolic health biomarkers in overweight/obese sedentary men. *Metabolism* 2013;62:212–9.
- 104 Fuhr U, Hellmich M. Channeling the flood of meta-analyses. *Eur J Clin Pharmacol* 2015;71:645–7.
- 105 Field AP. Dread returns to Mega-Silly One. *Health Psychol Rev* 2015;9:15–20.
- 106 Grindem H, Mansournia MA, Øiestad BE, *et al.* Was it a good idea to combine the studies? Why clinicians should care about heterogeneity when making decisions based on systematic reviews. *Br J Sports Med* 2018 bjsports-2018-099516–2012.